Introduction to Programming

Introductory study guide · Prepared as sample educational content for the StudentHub frontend project.

1. Programs and instructions

A program is a set of instructions a computer executes. Programming involves translating a problem into precise steps, expressing those steps in a language, and testing whether the result is correct.

2. Variables and data types

Variables are named references to values. Common data types include integers, floating-point numbers, strings, and booleans. Choose types that reflect the meaning of the data.

3. Input, processing, and output

Many programs follow an input-process-output pattern. A program may read a user's age, calculate an eligibility result, and display a message. Clear prompts and useful output make programs easier to use.

4. Decisions and repetition

Conditionals select actions based on a true/false test. Loops repeat a block of instructions while a condition holds or for each item in a collection. Check loop conditions carefully to avoid infinite loops.

5. Functions

A function groups a task under a name. It can accept parameters and return a result. Small functions with clear responsibilities make code easier to test, reuse, and explain.

6. Debugging and testing

Read error messages, reproduce the problem, isolate the relevant code, and test a small change. Test typical inputs, boundary values, and invalid input. A successful run does not prove every case is correct.

Pseudocode exercise

Ask for two numbers. Add them. Display the result. Extend the algorithm to reject input that is not numeric and explain how you would test it.